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MASTER'S INTERNSHIP PROPOSAL · 2027

Quantum key leasing for lattice-based signatures: towards concrete schemes

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LOCATION	PQShield, Paris (WeWork Bercy, 8 rue des Pirogues de Bercy, 75012)
DURATION	4–6 months.
TO APPLY	Send a CV, transcripts since L3, a short motivation statement and prior internship reports (if applicable) to damien.stehle@pqshield.com and thomas.espitau@pqshield.com

Context

Classical secret keys can be copied: once a signing key has been handed out, it can never truly be taken back. *Secure key leasing* (SKL) uses the no-cloning principle to fix this. The lessor hands out the secret key as a quantum state and can later obtain a certificate, ideally classical, proving that the lessee has irreversibly lost the ability to use it [AKN+23, APV23, CGJL25]. For signatures, this enables cryptographically enforced delegation: an employee, a device or a cloud service can be granted temporary signing rights, which are revoked when the key is returned.

Signature schemes with SKL have recently been built from lattice assumptions [MPY24, KMY25, KNP26, KLYY26], with classical revocation and even with a fully classical lessor. These constructions, however, target feasibility: they rely on unstructured lattices (SIS/LWE), may require the lessee to hold large quantum states and to perform substantial coherent computations, and come without concrete parameters. How far they are from what a realistic quantum device could run is unclear.

Objectives

1. **Understand the landscape.** Study the existing SKL signature schemes and identify, for each of them, which steps (key generation, signing, deletion, revocation check) genuinely require quantum computation and which can be made classical.
2. **Use rings.** Instantiate the most promising construction over polynomial rings or modules (Module-SIS and LWE [LS15]), as in ML-DSA, to shrink keys and quantum states, and check that the security arguments (certified deletion, extraction) carry over to the structured setting.
3. **Minimize quantum work.** Redesign the scheme so that the lessee's quantum operations (number of qubits held, quantum gates per signature) are as few as possible, ideally limited to preparing and measuring simple states, with the lattice arithmetic done classically.

4. **Set concrete parameters.** Derive parameters for standard security levels using lattice estimators, report qubit counts, key and signature sizes and quantum costs, and compare with the NIST post-quantum signature standards and state-of-the-art constructions.

Profile

Master's student (M2) in cryptography, theoretical computer science or quantum information, interested in both lattices and quantum computing. Familiarity with lattice-based cryptography or with basic quantum information (qubits, measurements, BB84 states) is expected; both are a plus.

References

[AKN+23] S. Agrawal, F. Kitagawa, R. Nishimaki, S. Yamada, T. Yamakawa. Public key encryption with secure key leasing. *EUROCRYPT 2023*.

[APV23] P. Ananth, A. Poremba, V. Vaikuntanathan. Revocable cryptography from learning with errors. *TCC 2023*.

[CGJL25] O. Chardouvelis, V. Goyal, A. Jain, J. Liu. Quantum key leasing for PKE and FHE with a classical lessor. *EUROCRYPT 2025*.

[KLYY26] F. Kitagawa, J. Liu, S. Yamada, T. Yamakawa. A unified approach to quantum key leasing with a classical lessor. *CRYPTO 2026*.

[KMY25] F. Kitagawa, T. Morimae, T. Yamakawa. A simple framework for secure key leasing. *EUROCRYPT 2025*.

[KNP26] F. Kitagawa, R. Nishimaki, N. Pappu. Collusion-resistant quantum secure key leasing beyond decryption. *EUROCRYPT 2026*.

[LS15] A. Langlois, D. Stehlé. Worst-case to average-case reductions for module lattices. *Designs, Codes and Cryptography*, 2015.

[MPY24] T. Morimae, A. Poremba, T. Yamakawa. Revocable quantum digital signatures. *TQC 2024*.